What changes an asteroid’s path?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how gravity can change velocity without contact.

SUCCESS CRITERIA - I can...

1. Draw velocity tangent to a path and force toward the attracting body.
2. Distinguish motion from a change in motion.
3. State what a straight-line model leaves out.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “acceleration” to explain your model or result. Sentence starters are optional.

gravity: an attractive interaction between masses

acceleration: rate of change of velocity

model: a drawing, explanation or equation for a physical situation

01 Warm-up: bring back an earlier idea

A 3 kg cart starts from rest under 15 N net force right for 2 s.
Find acceleration and final velocity.
State one assumption.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

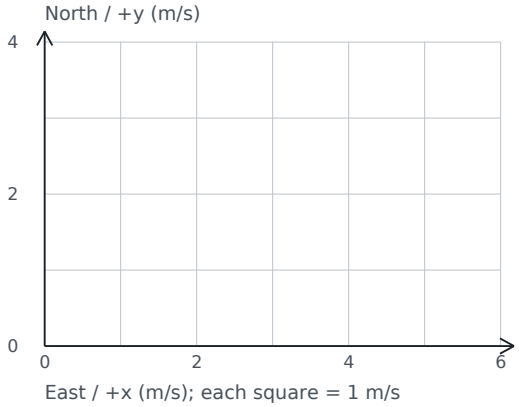
02 Worked example: explain the model

Reference: $a = F_{net}/m$.
Gravity attracts masses and can change speed, direction or both. Our asteroid scenario is fictional.

Worked reasoning: A fictional asteroid has a velocity arrow pointing right. A brief upward force gives it an upward change in velocity. The new velocity combines the original rightward motion with this change.

Your turn: A spacecraft is initially moving right. A short thruster burn adds an upward change in velocity.
Draw the original velocity and the change head-to-tail.
Draw the final velocity from the starting tail to the ending tip.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

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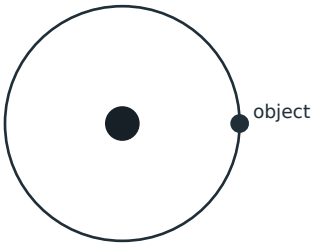
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch the velocity and gravitational-force arrows at the marked point on the orbit. Why do they point in different directions?

Sentence starter: Both … . They differ in … because … .



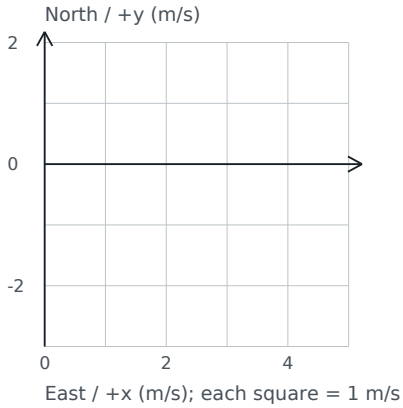
The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

In a flight simulator, a probe moves right at 4 m/s. After a 1 s thruster burn, it also moves down at 1 m/s. Add the components and final velocity to the grid.
Explain why the probe still moves right.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

05 Explain your reasoning

A student says space has no gravity because astronauts float.
Explain how orbiting astronauts can be falling.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

With a fixed net force, a 5 kg system accelerates at 14 m/s².
Predict acceleration for a 10 kg system.
State a control needed for this comparison.
Use the table to keep the controlled and changed quantities distinct.

Word help: system = the object or objects chosen for analysis

Sentence starter: I predict … . My model supports this because … .

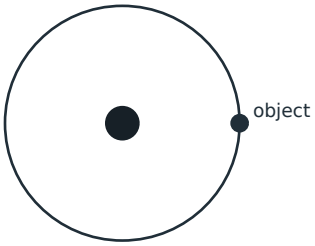
Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

07 Exit check

Name one observation that shows velocity is changing even if speed stays the same.

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A probe repeats the same thruster test, first with a short burn and then with a longer burn. Mass and thrust stay fixed.
Sketch a velocity-change arrow for each trial. Which burn causes a larger change, and why?

Sentence starter: Both … . They differ in … because … .

Compare the two burns

Right is positive.



Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3